

Masaki Imagawa

Curriculum Vitae

Last updated: October 2026

Contact Information

Institute of Mathematics, Academia Sinica
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Academic Appointments

Postdoctoral Fellow Institute of Mathematics, Academia Sinica, Taipei, Taiwan	Aug.2025–Present
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Education

Ph.D. in Informatics , Kyoto University Dissertation: Convergence analysis of elliptic regularization for advection equations in a bounded domain Advisor: Hiroshi Fujiwara	Apr.2022–Jul. 2025
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Master of Informatics , Kyoto University	Apr. 2020–Mar. 2022
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B.Sc. in Mathematics , Kyoto University	Apr. 2016–Mar. 2020
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Research Interests

Partial Differential Equations; Numerical Analysis; Conservation Laws; Viscous Shock Waves; Boundary Layers

Fellowships & Awards

SPRING program research fellow Division of Graduate Studies, Kyoto University / Japan Science and Technology Agency (JST)	Apr. 2022–Mar. 2025
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Publications

Journal Articles

Masaki Imagawa and Daisuke Kawagoe, A revisit on well-posedness of a boundary value problem of a stationary advection equation without the separation condition, *J. Math. Anal. Appl.* **551** (2025), no. 2, part 2, Art. 129695.
[doi:10.1016/j.jmaa.2025.129695](https://doi.org/10.1016/j.jmaa.2025.129695)

Conference Proceedings

Masaki Imagawa and Yuusuke Iso, Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain, in *2025 Days on Diffraction*, IEEE, 2025, 90–95.
[doi:10.1109/DD66835.2025.11263497](https://doi.org/10.1109/DD66835.2025.11263497)

Preprints

Masaki Imagawa and Daisuke Kawagoe,
On strong convergence of an elliptic regularization with the Neumann boundary condition applied to a stationary advection equation, [arXiv:2303.17904](https://arxiv.org/abs/2303.17904), 2023.

Presentations

Talks

Shock Waves and Boundary Layers for the Viscous Burgers Equation National Chung Hsing University Applied Mathematics Seminar	Oct. 2026 Taichung, Taiwan
Stability and Boundary Interaction of Viscous Shock Waves for the Burgers Equation 19th EASIAM Annual Meeting	Aug. 2026 Jeju, Korea
Stability of Shock Waves for the Viscous Burgers Equation with Boundary Interactions Taipei Postdoc Seminar, National Center for Theoretical Sciences	Apr. 2026 Taipei, Taiwan
Stability of Viscous Shock Waves for the Burgers Equation in a Half-Line Computational Mechanics Seminar, Kyoto University	Mar. 2026 Kyoto, Japan
Asymptotic stability of traveling waves for scalar viscous conservation laws with boundary Applied Mathematics Freshman Seminar 2025, Kyoto University	Nov. 2025 Kyoto, Japan
Convergence estimates for singular perturbation of advection equations with mixed boundary conditions Kyoto University Applied Mathematics Seminar	Jul. 2025 Kyoto, Japan
Convergence analysis of perturbed advection equations in a bounded domain Numerical Analysis Seminar, The University of Tokyo	Jul. 2025 Tokyo, Japan
Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain Days on Diffraction	Jun. 2025 Online
Boundary value problems of advection equations and its regularized problems Analysis Seminar, Academia Sinica	Mar. 2025 Online
On a Boundary Value Problem of a First-Order PDE and Its Regularization Problems 2024 Japan-Taiwan Joint Workshop on Numerical Analysis and Inverse Problems Kyoto University	Mar. 2024 Kyoto, Japan
Convergence Rates of Elliptic/Parabolic Regularization Applied to Advection Equations with Boundary Conditions The 5th Workshop on Recent Development of Mathematical Fluid Dynamics and Hyperbolic Conservation Laws	Sep. 2023 Jeju, Korea
On the Optimal Convergence Rate of Elliptic Regularization Methods Applied to Advection Equations The 19th Joint Meeting of JSIAM Activity Groups, Okayama University of Science	Mar. 2023 Okayama, Japan
On Convergence Rates of an Elliptic Regularization Applied to a Stationary Advection Equation The 13th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics National Center for Theoretical Sciences	Mar. 2023 Online
On Convergence Estimation of Elliptic Regularization Applied to Boundary Value Problem of Advection Equations 2022 Japan-Taiwan Joint Online Workshop on Numerical Analysis and Inverse Problem	Mar. 2022 Online
Numerical Computation of Advection Equations Using Elliptic Regularization Annual Meeting of the Japan Society for Industrial and Applied Mathematics	Sep. 2021 Online

Posters

Convergence rates of elliptic regularization applied to advection equations with boundary conditions
NCTS–Kyoto Mathematics Symposium, National Center for Theoretical Sciences

Dec. 2023
Taipei, Taiwan

On the Convergence Rate of Singularly Perturbed Solutions to the Steady Advection Equation
The 17th ICT Innovation, Kyoto University

Feb. 2023
Kyoto, Japan

Teaching Experience

Teaching Assistant, Kyoto University
Seminar on Applied Analysis I (Spring–Fall 2021)
Seminar on Applied Analysis II (Spring–Fall 2022)
Seminar on Nonlinear Mathematics (Fall 2023, Fall 2024)
Special Study Course (Mathematical Science) (Spring–Fall 2022)
Nonlinear Mathematics (Spring 2021, Spring 2022, Spring 2023)
Foundations of Numerical Computation (Spring 2021, Spring 2022, Spring 2023)
Function Theory on a Complex Variable (Fall 2021)
Advanced Calculus I (Spring 2020, Fall 2020)

Apr. 2020–Jan. 2024

Outreach

Breakout Session Instructor, Kyoto University Campus Guide
Research outreach program for high school students, organized in collaboration with the Osaka Prefectural Board of Education

Dec. 2024

Poster Session Advisor, Osaka Science Day
Provided feedback on high school students’ inquiry-based research presentations.

Oct. 2024

Outreach Lecturer, Kyoto University Learning Coordinator Initiative
Developed and delivered outreach lectures on mathematical modeling to high school students.
Tosajuku Senior High School
Fukui Prefectural Fujishima High School
Shanghai Japanese School Senior High School (Online)

Dec. 2024
Nov. 2024
Nov. 2023

Student Research Mentor, Kyoto Municipal Horikawa High School
Mentored high school students in inquiry-based research projects.

Apr. 2023–Sep. 2024

Qualifications

Specialized Teaching Certificates in Mathematics for Lower and Upper Secondary Schools, Japan

Mar. 2022